



AIRLINE CUSTOMER LOYALTY PROGRAM ANALYSIS

Data Warehouse & Business Intelligence Documentation

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1. Data Overview

This project analyzes an airline's customer loyalty program using two flat CSV files as source data. The goal is to build a complete data warehouse pipeline — from raw CSV files all the way to interactive Power BI dashboards — so airline management can track flight behavior, loyalty points, and customer lifetime value.

Source Files

File 1: Customer_Flight_Activity.csv

This file records each customer's monthly flight activity — it is the transactional source that becomes the Fact Table. It contains the Loyalty Number (customer ID), Year and Month of the activity, Total Flights taken, Distance flown in km, Points Accumulated that month, Points Redeemed, and the Dollar Cost of redeemed points.

File 2: Customer_Loyalty_History.csv

This file records the customer profile and loyalty program membership — it becomes the customer dimension source. It contains the Loyalty Number as the unique key, geographic details (Country, Province, City, Postal Code), demographics (Gender, Education, Salary, Marital Status), the customer's Loyalty Card tier (Star / Nova / Aurora), CLV (Customer Lifetime Value), Enrollment Type and date, and Cancellation date if the customer left the program.

2. Data Profiling

Before loading any data, we examine (profile) both files to understand the quality, structure, and characteristics of the data. This step helps us identify problems before they reach the database.

Profiling: Customer_Flight_Activity.csv

The flight activity file contains approximately 405,624 rows of monthly records. The Loyalty Number column is the key link to the loyalty history file and must be validated to ensure no orphan records exist. Data covers the years 2017 and 2018 across all 12 months. Some rows show zero flights and zero distance — these represent months where a customer was enrolled but did not fly, which is expected behavior. Many rows also show zero Points Redeemed since customers tend to accumulate over time and redeem infrequently. The most important issue found during profiling is that the date information is split across two separate integer columns (Year and Month) rather than a single date column — this must be resolved during transformation.

Profiling: Customer_Loyalty_History.csv

The loyalty history file contains approximately 16,737 unique customers. The Loyalty Number is the primary key and has no duplicates. Some customers have no Salary value recorded — these are kept as NULL in the Data Warehouse. The Cancellation Year and Month columns are empty for customers who are still active, which is expected. The Loyalty Card column has exactly three distinct values: Star, Nova, and Aurora. The Enrollment Type has two values: Standard and 2018 Promotion. The CLV field is a decimal number representing the predicted dollar revenue from each customer over their lifetime.

3. Logical & Physical Model

Before building the database, we design three levels of data models. Each level adds more technical detail.

3.1 Conceptual Model

The conceptual model describes the high-level business entities and how they relate to each other — in plain English, without any technical database details. It answers the question: "What are the main things we want to track?"

3.2 Logical Model

The logical model defines the five tables in the star schema and how they relate to each other. The central Fact Table connects to four dimension tables via foreign keys.

DimCustomer holds all customer attributes: Loyalty Number (primary key), Gender, Education, Salary, Marital Status, CLV, Enrollment Type, and an IsActive flag derived from whether a cancellation date exists.

DimLoyaltyCard holds the three loyalty tier types — Star, Nova, and Aurora — each assigned a surrogate CardID as the primary key.

DimLocation holds the unique geographic combinations of City, Province, and Postal Code, each assigned a surrogate LocationID.

DimDate is built from the source Year and Month integers. The DateKey is calculated as $\text{Year} \times 100 + \text{Month}$ (for example, January 2018 becomes 201801). A FullDate column is set to the first day of the month, MonthName is the full text name, and Quarter is derived from the month number.

The Fact Table stores one row per customer per month, containing the foreign keys to all four dimensions plus the measurable values: Total Flights, Distance, Points Accumulated, Points Redeemed, Dollar Cost of Points Redeemed, and CLV.

3.3 Physical Model

The physical model is the actual implementation in SQL Server. The diagram below shows the star schema built in Power BI / the DWH database, with the Fact_Table at the center connected to four dimension tables.

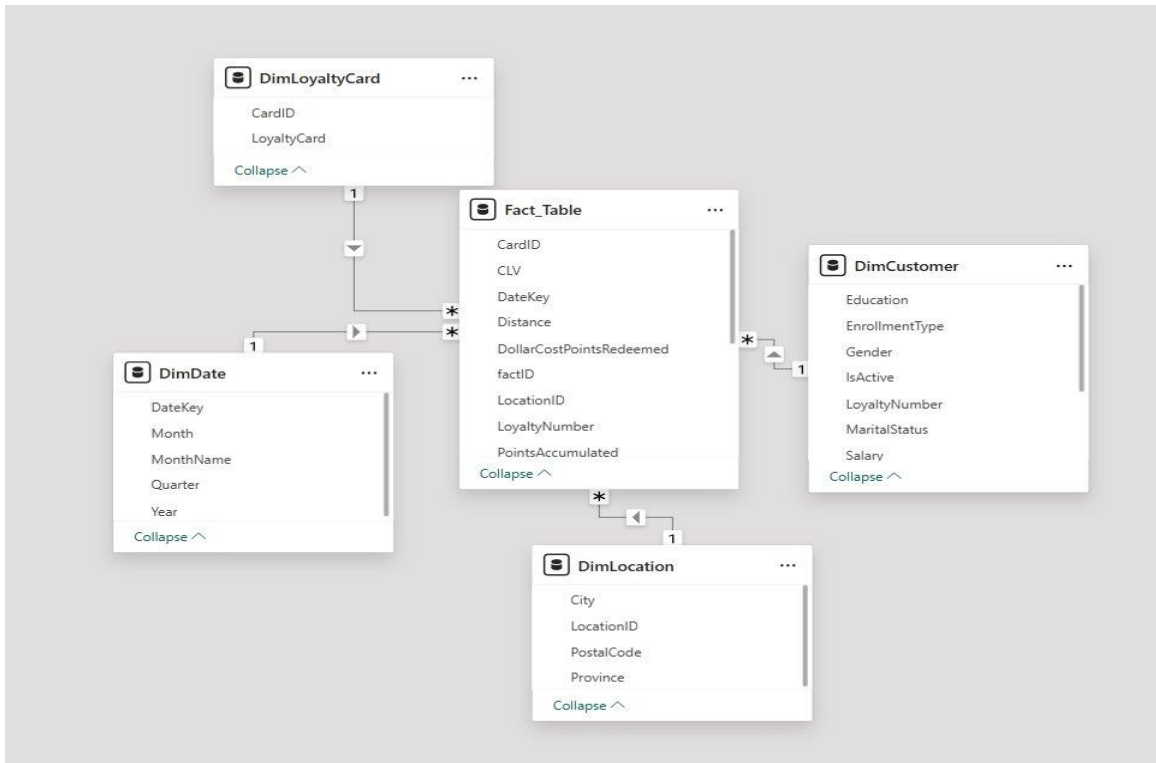


Figure 1: Star Schema — Physical Data Model (Power BI Semantic Layer)

The physical model implements a classic Star Schema. All dimension tables connect to the central Fact_Table via foreign key relationships. This design enables fast aggregation queries in Power BI.

One important transformation applied during physical loading: the Year and Month integer columns from the source CSV were converted into a proper DateKey (Year * 100 + Month), a FullDate column, a MonthName column, and a Quarter column — allowing time-based filtering and sorting in all visuals.

4. ETL Process

ETL stands for Extract, Transform, Load. It is the pipeline that moves data from the raw CSV files all the way to the final Data Warehouse. The pipeline uses SQL Server Integration Services (SSIS) and is organized into three database layers: ODS → STG → DWH.

4.1 ODS Layer — Load Raw Data from Flat Files

The first step in SSIS is to load both CSV files directly into the ODS database. Each file gets its own Data Flow Task. No transformation happens at this stage — the goal is simply to get the raw data into SQL Server as quickly as possible.

How it works in SSIS:

- A Truncate task runs first to clear any existing ODS data (full refresh approach).
- A Data Flow Task for Loyalty: Flat File Source reads Customer_Loyalty_History.csv → OLE DB Destination writes to ODS.Loyalty table.
- A Data Flow Task for Flight: Flat File Source reads Customer_Flight_Activity.csv → OLE DB Destination writes to ODS.Flight table.

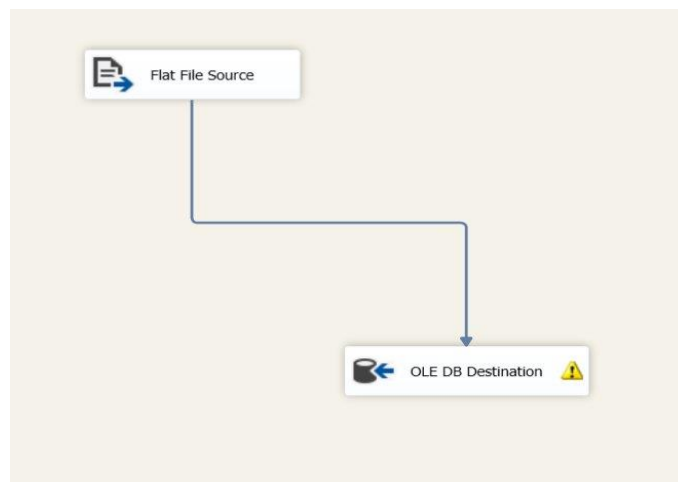


Figure 2: SSIS Data Flow — Flat File Source to OLE DB Destination (one task per CSV file)

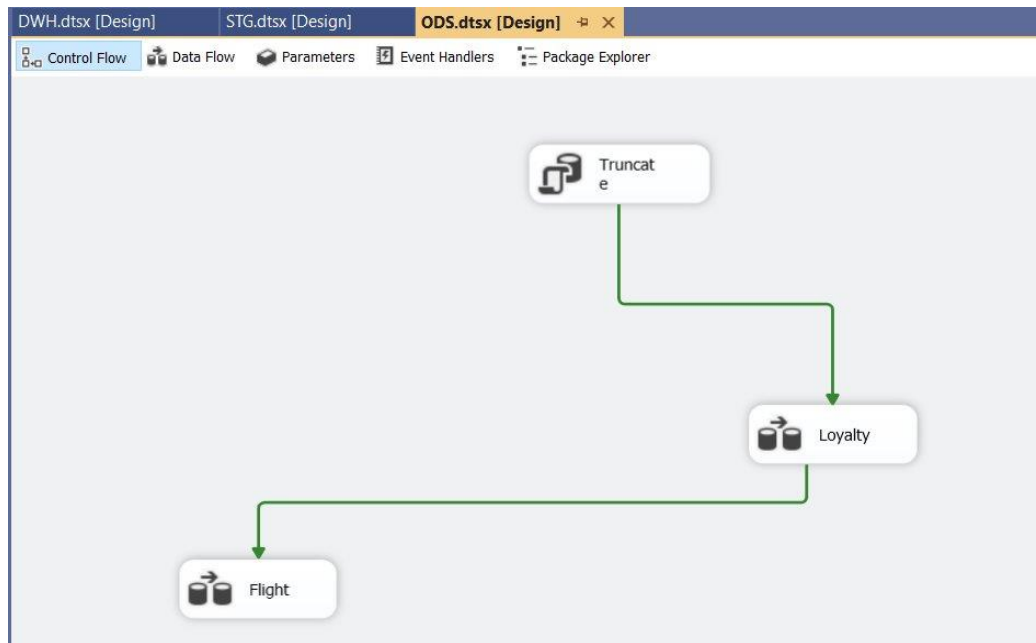


Figure 3: ODS.dtsx Control Flow — Truncate → Load Loyalty → Load Flight

4.2 STG Layer — Build Dimensions and OBT

Once the raw data is in ODS, the STG package transforms it into meaningful structures. A Sequence Container groups all steps and runs them in order.

Steps inside the STG Sequence Container:

- Truncate — clears all STG tables before reloading.
- OBT_Loyalty — creates a One Big Table that merges loyalty customer data with flight activity using a JOIN on Loyalty Number. This intermediate table simplifies all subsequent dimension extractions.
- OBT_Flight — prepares aggregated/cleansed flight data.
- Date — builds the DimDate table. Year and Month integers are combined: $\text{DateKey} = \text{Year} * 100 + \text{Month}$. FullDate is set to the first of each month, MonthName is derived using DATENAME(), and Quarter is DATEPART(QUARTER, FullDate).
- Customer — extracts distinct customer attributes from OBT_Loyalty and populates the staging customer table.
- LoyaltyCard — extracts the three distinct loyalty card types (Star, Nova, Aurora) and assigns surrogate keys.
- Location — extracts distinct City / Province / PostalCode combinations and assigns LocationID surrogate keys.

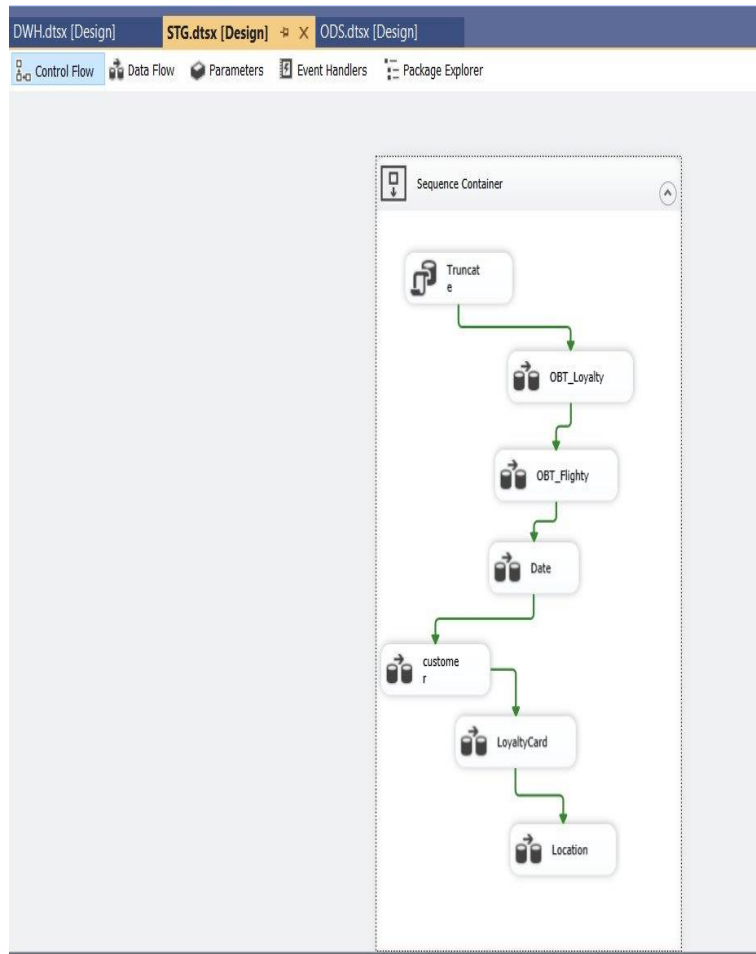


Figure 4: STG.dtsx — Sequence Container with all staging transformation steps

4.3 DWH Layer — Load Dimensions and Fact Table

The DWH package takes the clean STG data and loads it into the final star schema. It uses two Sequence Containers that run in order.

LoadDim Container — loads all dimension tables:

- Truncate → DimDate → DimCustomer → DimLocation → DimLoyaltyCard
- Each step reads from the corresponding STG table and inserts into the DWH dimension table.

LoadFact Container — loads the fact table:

- Truncate → Fact: reads from STG.OBT_Loyalty/OBT_Flight and resolves foreign keys by joining to the dimension tables (lookup on LoyaltyNumber → CustomerID, on card name → CardID, on location → LocationID, on Year+Month → DateKey).

- The Fact table is loaded only after all dimensions are fully populated, ensuring referential integrity.

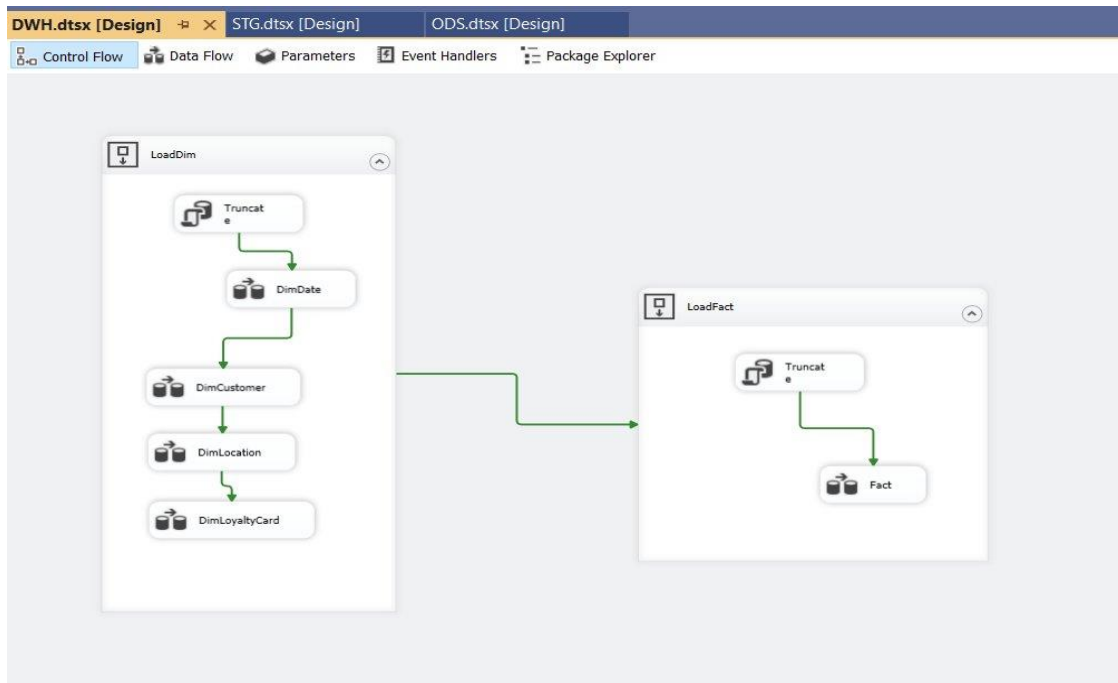


Figure 5: DWH.dtsx — LoadDim Container (left) followed by LoadFact Container (right)

5. Semantic Model

After the DWH is fully loaded, we connect Power BI Desktop to the SQL Server DWH database using Get Data → SQL Server. Power BI then automatically detects the relationships between tables based on matching key columns, forming the Semantic Model (also called the Data Model).

DAX Measures Used in the Report

DAX (Data Analysis Expressions) is the formula language used in Power BI to define calculated metrics. The key measures created for this project are:

Total Flights (SUM of all flights), Total Distance (SUM of all km flown), Total Points Accumulated (SUM of earned points), Total Points Redeemed (SUM of spent points), Average CLV (AVERAGE of Customer Lifetime Value across all customers), and Total Members (DISTINCTCOUNT of Loyalty Numbers to count unique enrolled customers).

These measures power all the KPI cards and charts in both report pages.

6. Power BI Overview

With the DWH fully loaded and the Semantic Model configured, the final step is building the Power BI report. The report contains two pages that present all key metrics and insights to airline management.

Page 1 — Executive Overview

This page gives a high-level summary of program performance at a glance. It is designed for executives and managers who need quick answers.

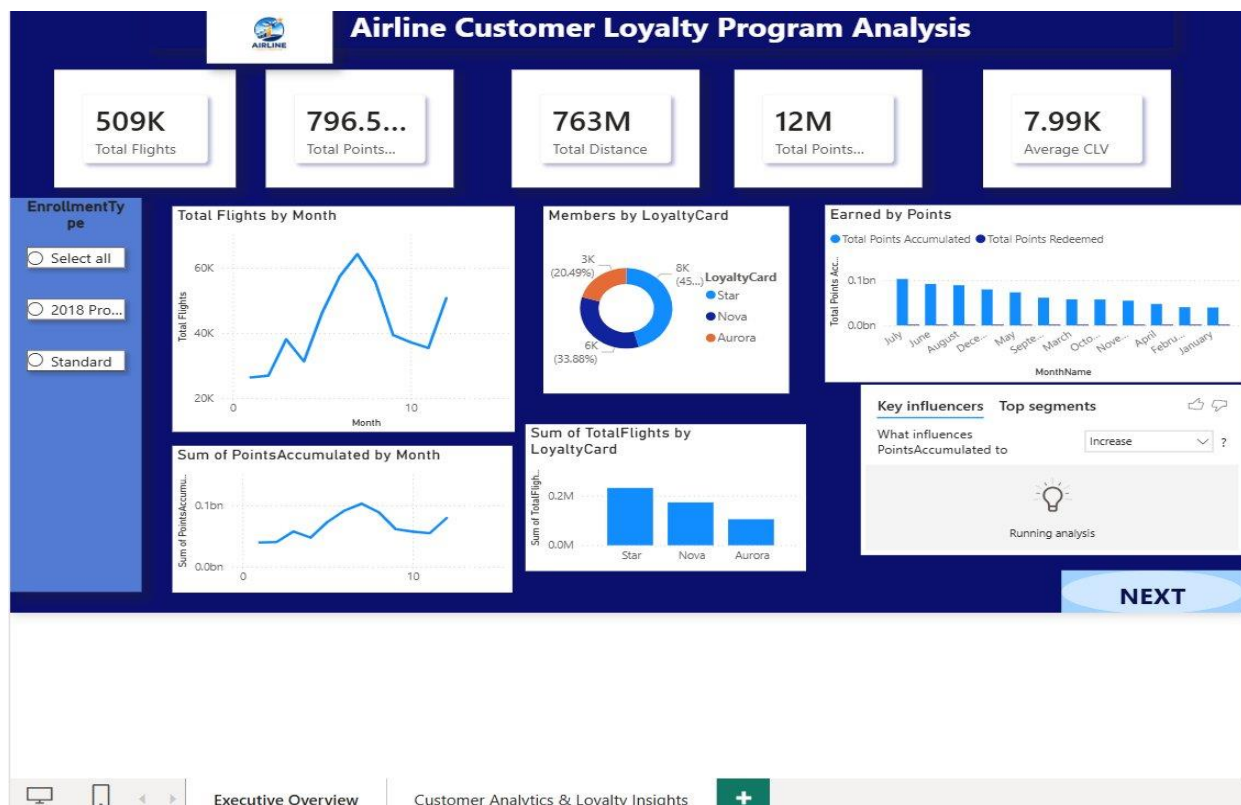


Figure 6: Power BI — Executive Overview Dashboard

Key Visuals on Page 1:

- KPI Cards (top row): Total Flights (509K), Total Points Accumulated (796.5M), Total Distance (763M km), Total Points Redeemed (12M), Average CLV (\$7,990).
- Total Flights by Month (line chart): shows seasonal patterns — a clear peak in summer months.
- Sum of Points Accumulated by Month (line chart): tracks loyalty engagement over time.

- Members by Loyalty Card (donut chart): Star = 45%, Nova = 34%, Aurora = 20%.
- Sum of Total Flights by Loyalty Card (bar chart): Star members fly the most.
- Earned by Points (clustered bar): compares Points Accumulated vs Points Redeemed per month — shows that customers earn far more than they spend.
- Enrollment Type slicer: allows filtering all visuals by Standard vs 2018 Promotion members.

Page 2 — Customer Analytics & Loyalty Insights

This page dives deeper into customer demographics and behavioral patterns across loyalty tiers.



Figure 7: Power BI — Customer Analytics & Loyalty Insights Dashboard

Key Visuals on Page 2:

- Customers by Education and Gender (bar chart): Bachelor degree holders are the largest group; males slightly outnumber females in most categories.
- Sum of CLV and Total Flights by Gender and Salary (scatter/bubble chart): high salary female customers tend to have the highest CLV.

- Total Distance by Loyalty Card (bar): Star members cover significantly more distance than Nova or Aurora.
- Customers by Marital Status (donut chart): Married = 58%, Single = 27%, Divorced = 15%.
- Total Flights by Month and Loyalty Card (line chart): all tiers decline slightly through the year, with Star consistently highest.
- Members by Year and Loyalty Card (stacked bar): growth across all tiers from 2017 to 2018.